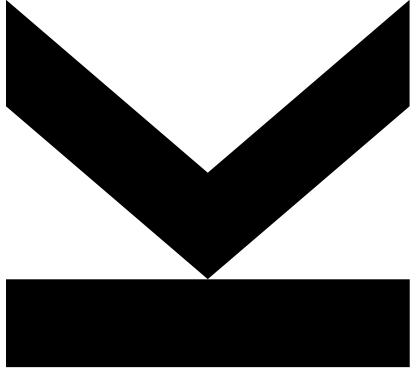


MLPC_2025_Task4_Challenge

Team Laborer



Ali Doğukan Seven, Karel Štícha, Yisong Tang, Dmitrii Troitskii

Evaluation Setup

- Stratified, file-level split: 70 % train / 15 % val / 15 % test: more data on training (Reduced base cost by 4%)
- Split by filename to avoid any temporal or label leakage
- Official metric = total cost per minute (segment level) — outweighs plain accuracy
- Naive majority-class baseline: 101.09 cost/min; top costly classes = Speech, Siren, Power Drill, Dog Bark (show table)

Total average cost per minute: 108.8553

	Avg. Cost per minute
Speech	24.8092
Shout	9.4118
Chainsaw	6.6057
Jackhammer	7.4642
Lawn Mower	7.7266
Power Drill	14.2369
Dog Bark	7.0986
Rooster Crow	0.3816
Horn Honk	12.7107
Siren	18.4102

Build a Simple SED system

- 120 ms frame embeddings → group 10 frames (1.2 s) and take the max probability
- Start with global threshold 0.5; later replace with per-class values
- No smoothing / median filtering applied
- Logistic-regression baseline ($C = 1$): 54.07 cost/min — already far below naive baseline

Total average cost per minute: 54.0684

	Avg. Cost per minute
Speech	6.3577
Shout	7.7138
Chainsaw	1.9555
Jackhammer	6.2432
Lawn Mower	5.2464
Power Drill	10.8458
Dog Bark	1.4992
Rooster Crow	0.2417
Horn Honk	8.2083
Siren	5.7568

Improvement Strategy #1 Hyper-parameter Tuning (Grid Search on C)

- Hypothesis: smaller C (stronger regularisation) curbs over-fitting
- Searched $C \in \{0.01, 0.1, 1, 10, 100\}$
- Best $C = 0.01 \rightarrow 50.34$ cost/min (-6.8 % vs 54.07)

C	total_cost
0.01	50.3383
0.1	51.0511
1	52.4959
10	52.7044
100	52.5129

C=0.01

Total average cost per minute: 50.3383

	Avg. Cost per minute
Speech	6.0683
Shout	7.2471
Chainsaw	3.5682
Jackhammer	4.0534
Lawn Mower	4.3342
Power Drill	9.7983
Dog Bark	1.6256
Rooster Crow	0.2766
Horn Honk	7.8961
Siren	5.4704

Improvement Strategy #2 Gaussian-Noise Data Augmentation

- Add Gaussian noise $\sigma = 0.02$ to each frame; triple the training samples
- Retrained under best-C = 0.01 setting
- Result: 50.57 cost/min — slight increase, no benefit
- For some classes the F1 score increases (highlighted)

C=0.01				
	BAcc	Precision	Recall	F1
Speech	0.902	0.657	0.855	0.743
Shout	0.797	0.233	0.625	0.34
Chainsaw	0.765	0.64	0.533	0.582
Jackhammer	0.799	0.348	0.608	0.443
Lawn Mower	0.79	0.384	0.588	0.464
Power Drill	0.774	0.282	0.572	0.378
Dog Bark	0.919	0.592	0.854	0.699
Rooster Crow	0.858	0.494	0.716	0.585
Horn Honk	0.783	0.197	0.589	0.295
Siren	0.862	0.617	0.732	0.67

data aug				
	BAcc	Precision	Recall	F1
Speech	0.899	0.654	0.849	0.739
Shout	0.786	0.224	0.604	0.327
Chainsaw	0.769	0.678	0.54	0.601
Jackhammer	0.781	0.412	0.568	0.478
Lawn Mower	0.745	0.381	0.498	0.432
Power Drill	0.767	0.289	0.557	0.38
Dog Bark	0.908	0.599	0.83	0.696
Rooster Crow	0.849	0.593	0.699	0.642
Horn Honk	0.776	0.197	0.574	0.294
Siren	0.856	0.643	0.721	0.68

data aug	
Total average cost per minute: 50.5702	
	Avg. Cost per minute
Speech	6.1003
Shout	7.6684
Chainsaw	3.4661
Jackhammer	3.8172
Lawn Mower	4.5768
Power Drill	9.6706
Dog Bark	1.6852
Rooster Crow	0.2575
Horn Honk	7.9088
Siren	5.4194

Improvement Strategy #3 Per-Class Threshold Tuning

- Grid-search thresholds 0.10 – 0.90 (step 0.05) per class
- Chosen thresholds minimise validation cost per label
- Total cost drops to 44.80 cost/min (-17.2 % vs baseline)

cost adjust

	BAcc	Precision	Recall	F1
Speech	0.899	0.653	0.85	0.739
Shout	0.737	0.313	0.49	0.382
Chainsaw	0.802	0.673	0.607	0.638
Jackhammer	0.749	0.556	0.501	0.527
Lawn Mower	0.772	0.418	0.55	0.475
Power Drill	0.734	0.337	0.484	0.397
Dog Bark	0.897	0.634	0.806	0.71
Rooster Crow	0.837	0.773	0.675	0.721
Horn Honk	0.753	0.304	0.518	0.383
Siren	0.836	0.748	0.676	0.71

adjust cost

Final LR (with per-class thresholds) val total cost: 44.7977

	Cost
Speech	6.12792
Shout	6.15771
Chainsaw	3.03204
Jackhammer	3.51717
Lawn Mower	4.14273
Power Drill	8.73867
Dog Bark	1.66816
Rooster Crow	0.185114
Horn Honk	6.00664
Siren	5.2215

Real-World Application Thinking

- Deployed model achieves 44.80 cost/min vs 54.07 baseline
- Use adaptive thresholds to cope with changing acoustics
- Design for incremental learning: plug-in new classes without full retrain; schedule periodic updates
- Cost-centred optimisation already realises >17 % operational savings

Team Laborer

Thank You